

Seokyoung Park

M.S. Student · Agricultural Robotics, Sensing, and Plant Phenotyping

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Research interests

My research focuses on image-based analysis and sensing for agricultural applications, with a particular interest in plant phenotyping. My current work includes hyperspectral assessment of tomato damage and spatial analysis of greenhouse environmental measurements. I also integrate sensors and data acquisition with robotic grippers and quadruped platforms. I aim to use imaging and sensing data to quantify crop traits and study their relationships with growing conditions.

Education

Pusan National University

Mar. 2025 – Feb. 2027 (expected)

M.S. in Bio-Industrial Machinery Engineering · GPA: 4.36/4.50

Pusan National University

Mar. 2017 – Feb. 2025

B.S. in Plant Bioscience · GPA: 4.26/4.50

Minor in Bio-Industrial Machinery Engineering

Summa Cum Laude; President's List.

Research experience

Graduate Researcher

Mar. 2025 – Present

Precision Agriculture and Smart Sensing System Lab., Pusan National University

Damage-aware robotic gripping for tomato harvesting

- Developed Python software for robotic compression experiments and force-signal processing using a myCobot 320 Pi arm, adaptive gripper and force sensor; compared four loading rates (6.9–15.3 N/s).
- Combined force–deformation analysis with hyperspectral measurements at 1 hour and 5 days after compression to study delayed damage; contributed to experimental design, analysis and manuscript preparation as first author.

Quadruped robotics for agricultural sensing

- Contributed to greenhouse microclimate mapping with a Unitree Go2 platform, measuring temperature, relative humidity and CO₂ at two canopy heights across eight lanes to produce 24 spatial maps and assess fixed-sensor representativeness. Co-first author of the greenhouse manuscript.
- Worked on reservoir upstream surveys using RGB imaging and RTK-GNSS to construct location-linked ground-image datasets. Plant phenotyping is a planned extension of the sensing platform.

Uplift resistance of expandable foundation piles

- Used Abaqus/Standard finite-element models to examine how wing inclination and dimensions influence pile uplift resistance in sandy soil; analysed load–displacement behaviour and soil plastic zones.

Load-cell sensing for air-supported domes

- Participated in load-cell installation and field acquisition at Damyang on 15 June 2026, recording cable-tension data through an indicator, USB-to-RS485 interface and Raspberry Pi.

Undergraduate Research Student

Mar. 2024 – May 2025

Functional Crop Laboratory, Pusan National University

- Worked on soybean leaf phenotyping under drought using time-series RGB images, YOLOv8, Roboflow and ImageJ, with Arduino-based environmental measurements.

Publications and manuscripts

Peer-reviewed publication

- [1] J. Park, S. Park, E. Jung, J. Song, G. Park, J. Lee, M. Kim, and J. Park. Inertial sensor applications in agriculture: Sensor types and data utilization techniques for mechanical and wearable applications. *Korean Journal of Agricultural Science*, 52(4):447–469, 2025. doi: 10.7744/kjoas.520405.

Under review

Park, S., Song, J., Hwang, H., Park, G., and Park, J. (2026). Speed-dependent control of robotic gripper for tomato harvesting to minimise compression-induced damage. *Biosystems Engineering*. **Under review**.

Manuscripts in preparation

Hwang, H.[†], **Park, S.**[†], and Park, J. Greenhouse microclimate mapping using a quadruped robot to assess fixed sensor representativeness. [†]Equal contribution.

Park, S., Park, J., Jang, J., Park, D., Kwon, S., Kim, M., and Choi, G. Analysis of uplift resistance characteristics of expandable piles in sandy soil with varying wing inclination and dimensions. *English translation of the Korean manuscript title*.

Research funding

- **Individual research grant:** National Research Foundation of Korea (NRF), funded by the Ministry of Education (MOE). Grant **RS-2025-25438074**.
Development of Locomotion Algorithms and Autonomous Control Systems for Quadruped Robots Specialized for Korean Agricultural Terrain.
- **Quadruped robotics:** National Research Foundation of Korea (NRF), funded by the Ministry of Science and ICT (MSIT). Grant **RS-2026-25475333**.
Development of an Integrated Sensing, Monitoring, and Decision-Making Model for Field Surveys in Agricultural Environments (Open Farmland and Greenhouses) Using a Quadruped Robot.
- **Air-dome foundations and sensing:** Korea Institute of Planning and Evaluation for Technology in Food, Agriculture and Forestry (IPET), through the Agri-Food Export Enhancement Technology Development Program, funded by the Ministry of Agriculture, Food and Rural Affairs (MAFRA). Grant **RS-2025-02313370**.
Development of Commercial-scale Vertical Farming System with Low Energy Consumption Based on Passive Air-dome Technology for Hot Climate in Middle East.

Technical skills

- **Programming and data analysis:** Python, NumPy, pandas and SciPy. Sensor time-series processing, force–deformation analysis, nonlinear model fitting and research visualisation.
- **Robotics and mobile sensing:** ROS 2, Unitree Go2, RTK-GNSS and LiDAR/IMU. Timestamped robot/sensor recording, greenhouse microclimate mapping, and matching ground images to GNSS positions and robot travel routes.
- **Instrumentation and data acquisition:** Force sensors, load cells, Raspberry Pi, Arduino and RS485. Force measurements for robotic compression tests, environmental measurements and field load-cell acquisition with CSV logging.
- **Imaging and plant phenotyping:** RGB and hyperspectral imaging; SAM, YOLOv8, Roboflow and ImageJ. Fruit-region segmentation and spectral comparison; soybean leaf-image analysis under drought stress and location-linked ground-image datasets.
- **Finite-element modelling:** Abaqus/CAE and Abaqus/Standard. Three-dimensional pile–soil contact models, displacement-controlled uplift simulations, geometric parameter studies and soil plastic-zone interpretation.

Selected conference presentations

- **Park, S.** (14 July 2026). Height-resolved 3D microclimate mapping and spatiotemporal analysis within greenhouse crop canopies using a quadruped robot. *AKABFE Student Research Spotlight, ASABE Annual International Meeting*, Indianapolis, USA. Presenter.
- **Park, S.,** Park, J., Hwang, H., and Kang, S. (2026). Ground-reaction-force-based lateral slope estimation for furrow-following quadruped robots. *Korean Society for Agricultural Machinery Spring Conference*.
- **Park, S.,** Park, J., Song, J., and Kim, M. (31 October 2025). Analysis of pile pull-out behaviour in Middle Eastern soil using digital image correlation. *Korean Society of Agricultural Engineers Autumn Conference*, Jeju.
- Hwang, H., **Park, S.,** et al. (22 May 2025). Remote control and integrated sensor-data environment for an agricultural quadruped robot. *Korean Society for Precision Agriculture Spring Conference*. Co-author.
- **Park, S.,** Park, J., Park, G., Song, J., and Hwang, H. (25 April 2025). Minimising damage and optimising control in fruit and vegetable harvesting using a robotic arm and gripper with real-time force sensing. *Korean Society for Agricultural Machinery Spring Conference*.

English translations are used for Korean presentation titles.

Honours and awards

- **Best Paper Presentation Award (team)**, Korean Society for Precision Agriculture, May 2025 and November 2024.
- **Excellent Paper Presentation Award**, Korean Society for Agricultural Machinery, April 2025 and October 2024.
- **Excellence Award**, PNU-PKNU Field Problem Research Group Final Symposium, November 2024.
- **Best Presentation Award (team)**, Korean Society for Engineering Education, September 2024.
- **Bronze Prize**, PNU College of Natural Resources and Life Science Capstone Design Competition, October 2023.
- **Excellence Award**, Busan LINC 3.0 Entrepreneurship Shared University Competition, January 2023.
- **PNU merit scholarships**: four full-tuition and two partial-tuition awards, 2017–2024.

Selected coursework

Graduate: Biological Image Processing; Statistical Processing and Data Analysis in Bio-Industrial Machinery Engineering; Advanced Bio-Industrial Machinery Design; Biosystems Mechanics; Advanced Biomaterials Engineering I.

Undergraduate – engineering and sensing: Biosensor and Bio-Instrumentation & Practice; Biosystems Control Engineering; Bio-Production Mechatronics Design; Applied Mechanics of Materials; Bio-Industrial Machine Element Design (I); Food Process Engineering & Practice.

Undergraduate – programming and quantitative methods: Basic Computer Programming; Experimental Design for Biologists; Biostatistics; Calculus (II).

Undergraduate – plant science and cultivation: Plant Physiology and Practice; Plant Genetics; Plant Molecular Biology; Plant Biotechnology; Plant Breeding (I, II); Soil Science; Plant Ecology; Protected Horticulture.

Additional training

Modeling Contact with Abaqus/Standard, VPK PLM Solution Training Center, 25–26 June 2026.

Introduction to Abaqus/CAE, VPK PLM Solution Training Center, 18–20 March 2026.